

INFO 2646 | INTRODUCTION TO DATA SCIENCE

Who Grooms Whom?

Finding social subgroups in the Edinburgh Zoo chimpanzee community with social network analysis.

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Final Project Pitch · Spring 2026

What I'm trying to do

17

chimpanzees observed

I want to figure out which chimps form social subgroups in this 17-individual community, using grooming behavior as the signal.

The Question

Who's grooming whom, and how tight are those bonds?

The Method

Turn raw grooming events into a sociomatrix, then run community detection on the resulting network.

The Payoff

Subgroup structure can inform husbandry decisions and flag any individuals on the edge of the network.

Where the data came from

Executive Summary

I collected this data at the Edinburgh Zoo over 39 observation days between April 10 and June 15, 2017 as a part of my master's thesis. Two observations ran concurrently: presence checks every two minutes (was each chimp visible?) and all instances of grooming and aggression events.

Data Source

Author: Joshua Moran (Me!)

Site: Edinburgh Zoo

Period: April 10 – June 15, 2017

Method: 2-min scan sampling and all-occurrences behavioral observations



Two tables, two units of observation

Presence	Behavioral Data
2979 19 <i>observations</i> <i>columns / factors</i> Was each chimp visible? Recorded as 1 or 0 every two minutes. Fields: Date, Time, plus 17 [Chimp]_Present columns. Unknown_Present is dropped.	437 10 <i>observations</i> <i>columns / factors</i> One row per interaction: who groomed (or aggressed) whom, when it started, when it ended, and how long it lasted. Fields: Date, Actor, Behavior, Receiver, Start, End, Duration, AM/PM, Groomer sex, Groomee sex

Half tidy, half wide

Presence — not tidy

Wide format. Each chimp gets its own column, which makes joining and grouping awkward.

Date	Time	Cindy	Edith	...
4/10	10:05	0	0	...
4/10	10:32	0	1	...
4/10	10:36	0	1	...

Reshape with `pivot_longer()` into `chimp_id` / `present` columns before joining with the grooming data.

Grooming Data — tidy

One row per event, one variable per column.

Date	Actor	Behavior	Receiver	Duration
4/10	Kindia	Mutual groom	Liberius	0:14:19
4/10	Paul	Allogroom	Liberius	0:04:41
4/10	Frek	Mutual groom	Qafzeh	0:08:38

Already in tidy form. Ready for sociomatrix construction.

Factors and types

Grooming Data: 10 factors / 437 observations. Presence: 19 factors / 2,979 observations.

Field	Source	Type	Example / Range
Date	Both	Numerical (date)	2017-04-10 → 2017-06-15
Time / Start / End	Both	Numerical (time)	10:32:28 · Start/End not used in modeling
Duration	Grooming	Numerical (right-skewed)	0 – 19.95 min · median 1.5 min
[Chimp]_Present	Presence	Categorical (binary 0/1)	0 = absent, 1 = present
Actor / Receiver	Grooming	Categorical (17 levels)	Cindy, David, ..., Velu
Behavior	Grooming	Categorical (3 levels)	Allogroom (350), Mutual groom (72), Aggression (15)
AM/PM	Grooming	Categorical (2 levels)	AM, PM
Groomer / Groomee sex	Grooming	Categorical (2 levels)	Male (9), Female (8)

Data is mostly clean, two adjustments needed

No missing or null values anywhere in the data.

1. Filter out aggression

15

aggression events to drop

Leaves 422 affiliative events: 350 allogrooms and 72 mutual grooms. Aggression is a separate question for a separate network.

2. Drop Unknown_Present Column

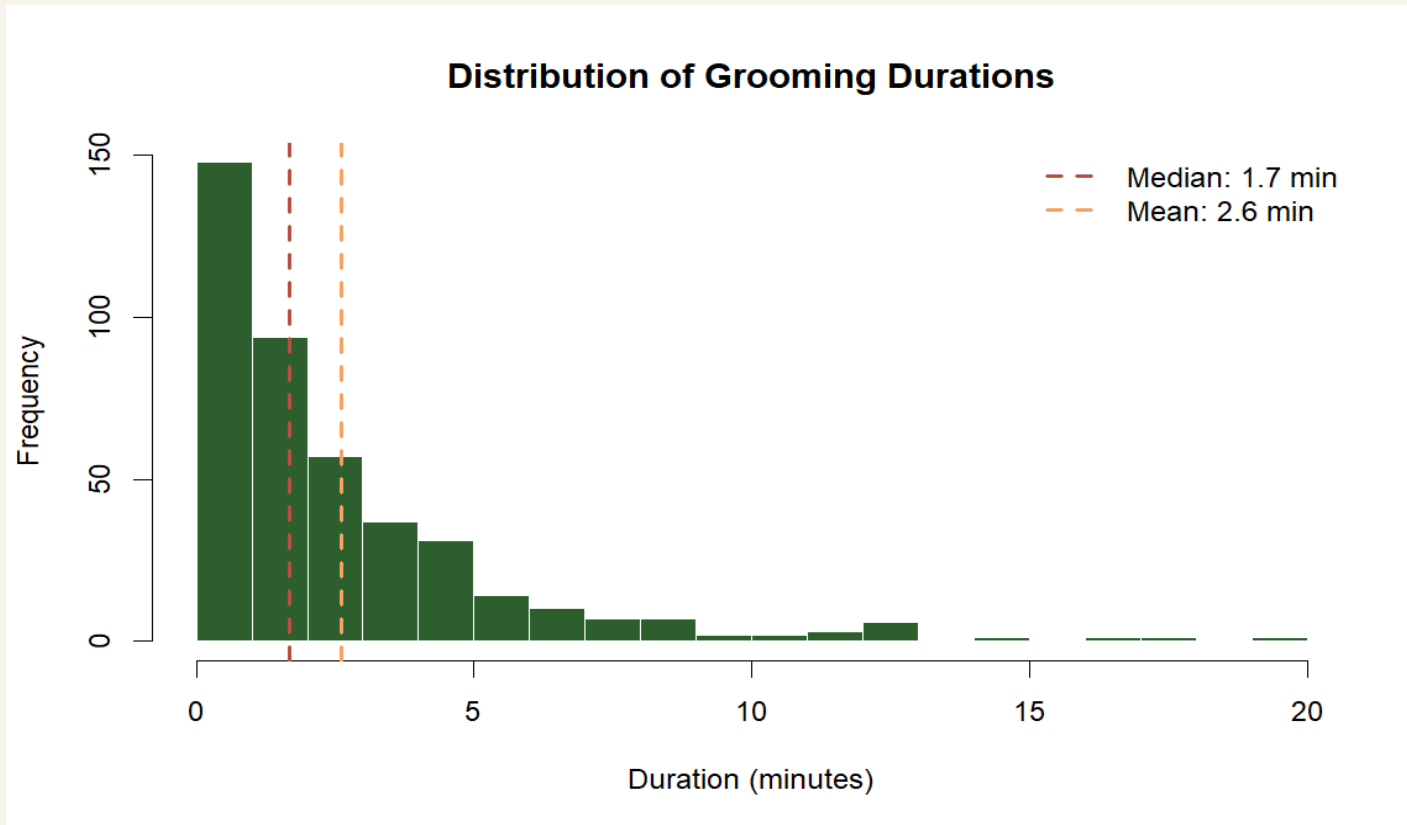
1

column to remove

Unknown_Present can't be tied to a specific chimp, so it adds nothing for subgroup detection. After dropping it, 17 named chimps remain plus Date and Time.

DISTRIBUTION

Most grooming bouts are short. A handful run long.



19.95

max minutes

1.7

median minutes

2.6

mean minutes

87%

of events under 5 minutes

If a method assumes normal weights, log-transform first.
Modularity-based community detection doesn't, so the raw values are fine.

Data munging time!

Step	Why it matters
Convert Duration to decimal hours	Stored as HH:MM:SS. Convert to decimal so durations are easy to sum and compare.
Drop the 15 aggression events	Aggression and grooming aren't the same network. The affiliative network keeps the 422 grooming rows.
Drop Unknown_Present	Can't tie it to a specific chimp, so it adds nothing and could distort sampling rates.
Reshape Presence to long form	<code>pivot_longer()</code> into <code>chimp_id</code> / <code>present</code> columns. Required for joint-presence calculations.
Correct for sampling effort	Divide grooming totals by minutes both chimps were present at the same time during observations. Without this, the most-observed pairs look closest by default.
Log-transform skewed weights	Duration is right-skewed. If the modeling step assumes normal weights, log-transform the edge weights first.
Watch for dyadic non-independence	Standard regression assumes independent rows. Dyads aren't independent. Use quadratic assignment procedure or permutation tests for inference.

A quick primer

What is it?

Social network analysis studies systems by their **relationships** rather than their attributes. Each entity (in this case a chimp) is a **node**, each relationship is an **edge**, and the result is a graph that can be measured.

The terminology is very similar to artificial neural networks, however ANNs emulate neurons that learn to recognize patterns in data through layers of weighted connections.

Origins

Came out of sociology in the 1930s with Moreno's sociograms. Now used in a variety of fields, including epidemiology, organizational research, and animal behavior.

What SNA can answer

1

Connection

Who interacts with whom, and how often?

2

Centrality

Who sits at the heart of the network? Who's on the edge?

3

Community

Are there natural clusters or cliques?

4

Diffusion

How do behaviors or information spread between nodes?

SNA with Louvain community detection

Louvain modularity

an unsupervised graph-clustering algorithm

Each chimp is a node. Each grooming interaction is a weighted edge. The Louvain algorithm splits the network into groups that maximize modularity (dense ties within groups, sparse ties between them).

Why this fits the data

- 1 Data is relational**
Every row already names two individuals. That's the shape of a graph, not a feature matrix.
- 2 No labels**
We don't know which chimps belong to which subgroup. SNA finds structure without that label.
- 3 Weights mean something**
Grooming duration is a real proxy for bond strength. Modularity uses those weights directly.
- 4 Standard in primatology**
Published chimpanzee studies (e.g., Lehmann & Boesch 2009) use exactly this approach.

What I'll be able to answer, and what to do with it

The main question

How many social subgroups are there in this 17-chimp community, and who's in each?

Secondary questions:

- Are subgroups segregated by sex?
- Which individuals are bridges between cliques?
- Do subgroups align with the 2 groups that were integrated to form this group?

Actions this drives

Welfare

Flag any chimp with very few connections so caretakers can check in.

Husbandry

Help with pairing and grouping decisions during enclosure rotations.

Research

Set up follow-up questions, like whether kinship predicts subgroup membership.

Comparison

Give a baseline to compare against later data, e.g. after an intervention or in a different season.

From raw events to a 17×17 weighted matrix

1

Filter

Keep allogroom and mutual groom rows. Drop the 15 aggression rows.

2

Aggregate

Sum Duration (sec) for every Actor $\rightarrow$ Receiver pair across the study period.

3

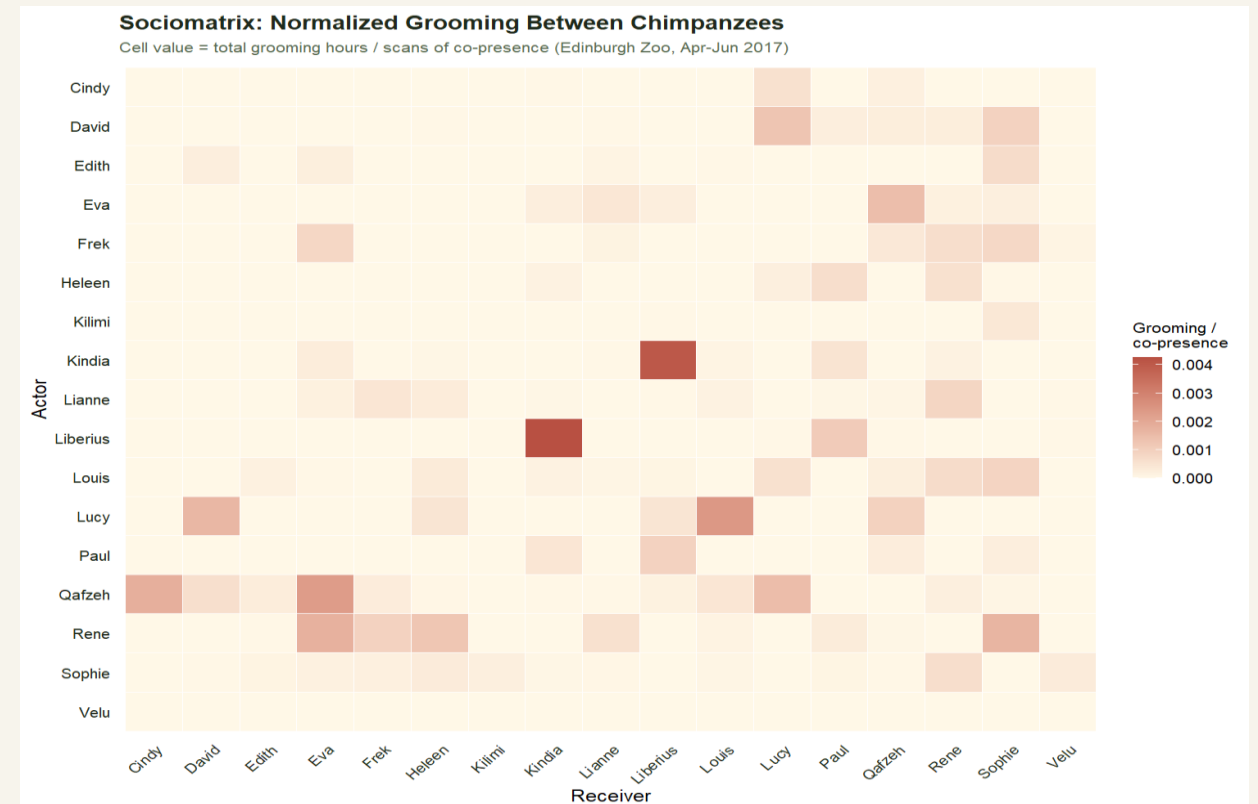
Symmetrize

Add mutual groom in both directions, then treat the network as undirected for community detection.

4

Normalize

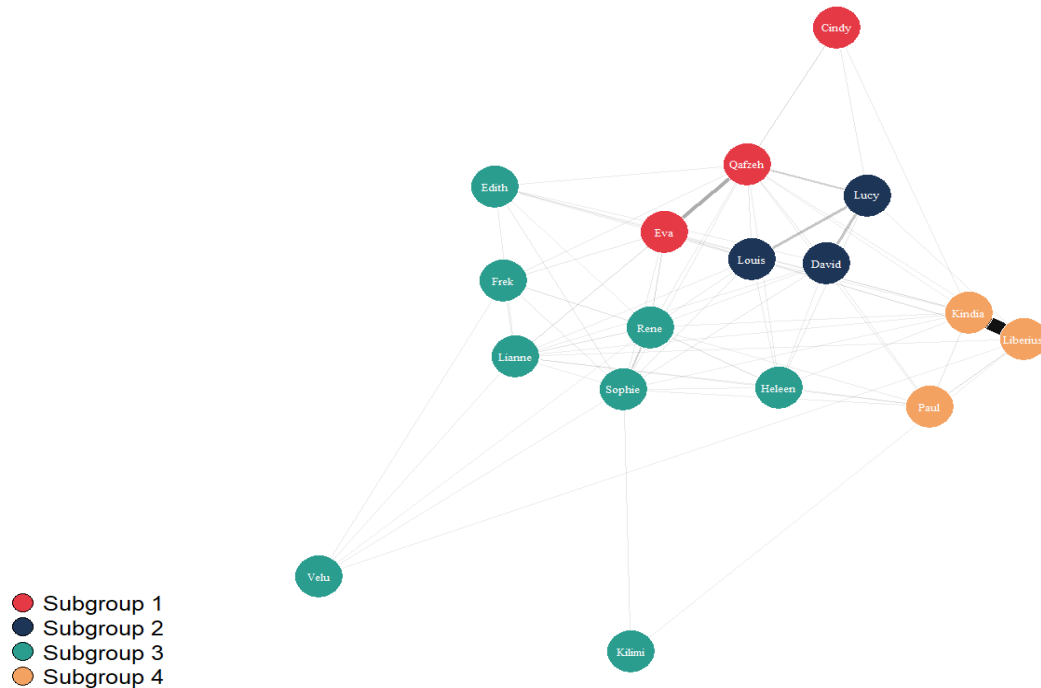
Divide by joint observation time so the most-observed pairs aren't artificially heaviest.



RESULT

Four subgroups detected in the grooming network

Detected Chimpanzee Subgroups via Louvain Community Detection



Subgroup 1

Cindy · Eva · Qafzeh

Small mixed-sex cluster anchored by the Eva ↔ Qafzeh dyad.



Subgroup 2

David · Louis · Lucy

Mixed-sex triad, densest tie Lucy ↔ Louis.



Subgroup 3

Edith · Frek · Heleen · Kilimi · Lianne · Rene · Sophie · Velu

The largest cluster — the social core, female-skewed (5F, 3M).



Subgroup 4

Kindia · Liberius · Paul

Tight all-male triad. Kindia ↔ Liberius is the heaviest single dyad.